

Enhanced Drone Navigation and the Integration of GANs in Kalman Filter-Based Target Pursuit

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Fig.1. Bebop 2 Drone prototype, equipped with Advanced face recognition technology integrating Thermal-to-Visible GANs and SiGAN for reliable verification in low-light environments, showcasing potential for rescue operations.

Abstract

This project presents significant advancements in drone navigation and target pursuit through the integration of Kalman filters for precise state estimation and prediction. The proposed system effectively addresses challenges such as low-light conditions and noisy sensor data, substantially enhancing drone performance in real-world applications like surveillance, search and rescue, and facial recognition. A key contribution of this work is the development of a target pursuit drone architecture that combines Kalman filter-based state estimation with advanced face recognition pre-processing techniques, including Thermal-to-Visible Generative Adversarial Networks

(GANs) and Siamese Network GANs (SiGAN). These innovations improve face detection and recognition accuracy in adverse environmental conditions.

Introduction

Drones have become indispensable tools in a variety of fields, ranging from recreational photography to critical operations such as search and rescue. However, their adoption has been hindered by persistent challenges, including navigation in environments characterized by noisy sensor data and low visibility. These challenges become even more pronounced in target pursuit and recognition tasks, where precision and reliability are essential. Kalman filters, known for their effectiveness in handling noisy measurements, provide an ideal framework for tackling these challenges [4].

This project focuses on two primary goals: enhancing the accuracy of drone navigation and target pursuit, and addressing the limitations of face detection and recognition under poor visibility and adverse conditions. By leveraging the predictive capabilities of Kalman filters, the system achieves a robust design capable of meeting the demands of real-world applications.

The project's development was carried out in several phases, beginning with a comprehensive literature review to identify existing gaps and opportunities. This was followed by careful system design and algorithm development, culminating in extensive real-world testing. The results demonstrate a highly reliable navigation system with improved accuracy, dependable target tracking, and optimized facial recognition performance, even under challenging conditions. By utilizing state-of-the-art methods and technologies, this project contributes significantly to the rapidly evolving field of drone technology and highlights its potential for critical real-world applications. The findings open pathways for innovative solutions in dynamic and complex operational scenarios, paving the way for the next generation of intelligent drones.

Related Works

Recent advancements in artificial intelligence and control systems have contributed significantly to enhancing drone technology, particularly in navigation and target pursuit. Various studies showcase key methodologies and frameworks that align with the topic of integrating Generative Adversarial Networks (GANs) with Kalman Filters for improved drone navigation.

Fernando's work on "Enhancing Perception for Autonomous Driving Systems with Dual-Vision Input Cycle GAN for Night-to-Day Image Translation" demonstrates the application of GANs in visual data processing. This study is pivotal for drone navigation in low-light environments, enabling enhanced perception and operational capabilities [1]. Similarly, Wang, Chen, and Wu's research on "Thermal to Visible Facial Image Translation Using GANs" focuses on bridging the modality gap between thermal and visible facial imagery, a technique crucial for target identification in poor visibility conditions [2].

Loquercio et al.'s paper, "DroNet: Learning to Fly by Driving," presents a data-driven navigation system that employs neural networks to adaptively improve drone navigation in dynamic and

unpredictable environments. This framework highlights the versatility of neural networks in guiding drones in real-world scenarios [3]. On the other hand, Manjunatha, Pak, Filev, and Tsiotras explore the integration of Kalman filters with recurrent neural networks (RNNs) in "KARNet: Kalman Filter Augmented Recurrent Neural Network for Learning World Models in Autonomous Driving Tasks." This study introduces an adaptable framework for drones to navigate and predict in complex settings effectively [4].

Expanding on Kalman filter applications, Kim, Petrunin, and Shin's review paper, "A Review of Kalman Filter with AI Techniques," categorizes methods for enhancing Kalman filters using artificial intelligence. These improvements address challenges like noisy data and dynamic conditions, a critical alignment with the goal of enhancing drone navigation precision [5]. Additionally, Poster et al.'s "A Large-Scale, Time-Synchronized Visible and Thermal Face Dataset" introduces a robust dataset for thermal-to-visible image translation. This resource is invaluable for advancing facial recognition under challenging environments, essential for accurate target pursuit tasks [6].

The adaptive capabilities of Kalman filters are further explored in the paper "An Adaptive Unscented Kalman Filter for Vehicle Dynamics." The study presents innovative methods for precise state estimation under varying environmental conditions, showcasing adaptability that could be applied to drone operations in complex scenarios [7]. Finally, Tahir, Mir, and Islam's review paper, "Control Algorithms, Kalman Estimation, and Near-Actual Simulation for UAVs: State of Art Perspective," provides foundational insights into control systems and Kalman filters for UAVs, emphasizing predictive capabilities essential for reliable drone navigation and pursuit [8].

These works collectively contribute to the evolving landscape of drone technology by offering methodologies that enhance navigation, recognition, and adaptability, paving the way for integrating GANs and Kalman filters in a unified drone system.

Integrated Advancements in Drone Technology

While existing studies often focus on the individual applications of Kalman filters or Generative Adversarial Networks (GANs), this project takes an innovative approach by integrating these two advanced technologies into a unified system. This novel integration sets the project apart by addressing broader challenges and enhancing the multifaceted performance of drones.

Dual Enhancement: Unlike the reviewed papers that target navigation or face detection independently, this project achieves a breakthrough by combining Kalman filter-based state estimation with GAN-powered preprocessing. This dual approach simultaneously enhances both navigation and facial recognition capabilities in drones, creating a more comprehensive solution for real-world applications.

Environmental Adaptability: Kalman filters are employed in this project to effectively tackle challenges like noisy sensor data and low-light conditions, ensuring precise state estimation. Complementing this, GANs optimize visual data processing, as exemplified by techniques like Thermal-to-Visible GANs. Together, these technologies offer superior adaptability, making the system highly effective for operations in challenging environments, such as surveillance or search-and-rescue missions.

Target Pursuit Architecture: A key innovation of the project is the incorporation of GAN techniques such as Siamese Network GANs (SiGAN), which significantly improve facial recognition and tracking. This addition provides a layer of predictive accuracy that surpasses conventional navigation systems, including those discussed in papers like KARNet or DroNet. By enhancing target pursuit capabilities, the project ensures optimal performance in adverse and dynamic conditions.

Unified System Design: While other studies explore standalone advancements—whether adaptive Kalman filters, GAN-based image translation, or neural network navigation—this project introduces a comprehensive architecture. By merging these technologies, it addresses broader challenges that drones face in dynamic and complex scenarios. This unified system design not only leverages the strengths of each technology but also creates synergies that significantly improve overall performance.

In summary, this project stands out from existing research by integrating Kalman filters and GANs into a single, robust system. It effectively combines precision state estimation, optimized visual processing, and advanced recognition capabilities, paving the way for a new generation of intelligent drones.

In essence, this project builds upon and bridges the strengths of existing research, creating a system that transcends their individual contributions to achieve enhanced functionality in drone navigation and target pursuit.

Kalman Filters for Drone Navigation

Kalman filters have emerged as essential tools in drone navigation for their ability to handle noisy measurements and improve tracking accuracy. Research has demonstrated the effectiveness of these filters in mitigating uncertainties inherent in dynamic environments. Studies have also explored the combination of artificial intelligence techniques with Kalman filters for drone detection and tracking, highlighting the adaptability of these filters in handling cluttered and noisy conditions.

GANs for Face Detection in Low-Light Conditions

Generative adversarial networks (GANs) have demonstrated remarkable potential in enhancing face detection accuracy under adverse conditions such as low light. Thermal-to-visible GANs represent a significant breakthrough in this field, enabling face detection by generating high-quality visible images from thermal inputs. Additionally, Siamese Network GANs (SiGANs) have gained prominence for their efficacy in face verification tasks, combining Siamese networks with GANs to reconstruct high-resolution faces while preserving identity.

Integration of Kalman Filters and GANs

While existing studies focus on Kalman filters and GANs separately, this project uniquely integrates these technologies to enhance both navigation and face detection capabilities for drones. By applying thermal-to-visible GANs, the project successfully addresses low-light conditions, demonstrating effectiveness in practical use cases. The incorporation of SiGAN for face recognition adds robustness to the detection process, enabling drones to identify individuals with greater accuracy.

Dataset

To facilitate accurate navigation and face recognition, this project utilized various datasets to train and validate the models.

Steering Angle Dataset

To learn steering angles from images, the project employed a publicly available dataset from **Udacity's autonomous driving project**. This dataset consists of over **70,000 images of car driving** distributed across **six experiments**, with five used for training and one for testing. Each experiment contains time-stamped images captured from three different cameras (left, central, and right), along with **IMU, GPS data, gear, brake, throttle, steering angles, and speed**. For this implementation, only images from the **forward-looking camera** were used alongside their associated steering angles.

Collision Dataset

To improve obstacle avoidance mechanisms in drone navigation, a dedicated **collision dataset** was employed. This dataset was gathered by recording video sequences where the drone moves toward obstacles. The recording begins when the drone is far from an obstacle and stops when it gets very close. The dataset contains **approximately 32,000 images distributed over 137 sequences**, capturing diverse obstacle types. Frames are manually annotated, with those far from collisions labeled **as 0 (no collision)** and frames near an obstacle labeled **as 1 (collision)**.

Face Recognition Dataset

For face detection and recognition tasks, this project utilized two key datasets:

- **IRIS Dataset:** A widely recognized dataset for identity verification.
- **Kaggle Face Dataset:** Available at this link <https://www.kaggle.com/datasets/rajatdulal/face-dataset>, this dataset contains a large number of annotated facial images suitable for training face recognition models.

The combination of **thermal-to-visible GANs and SiGAN-based facial recognition algorithms** was trained and validated using these datasets, ensuring robust performance in various lighting conditions.

Drone Target Pursuit integrates Drone Navigation with advanced Drone Face Recognition technology.

Drone Navigation

Bebop Drone Model Training

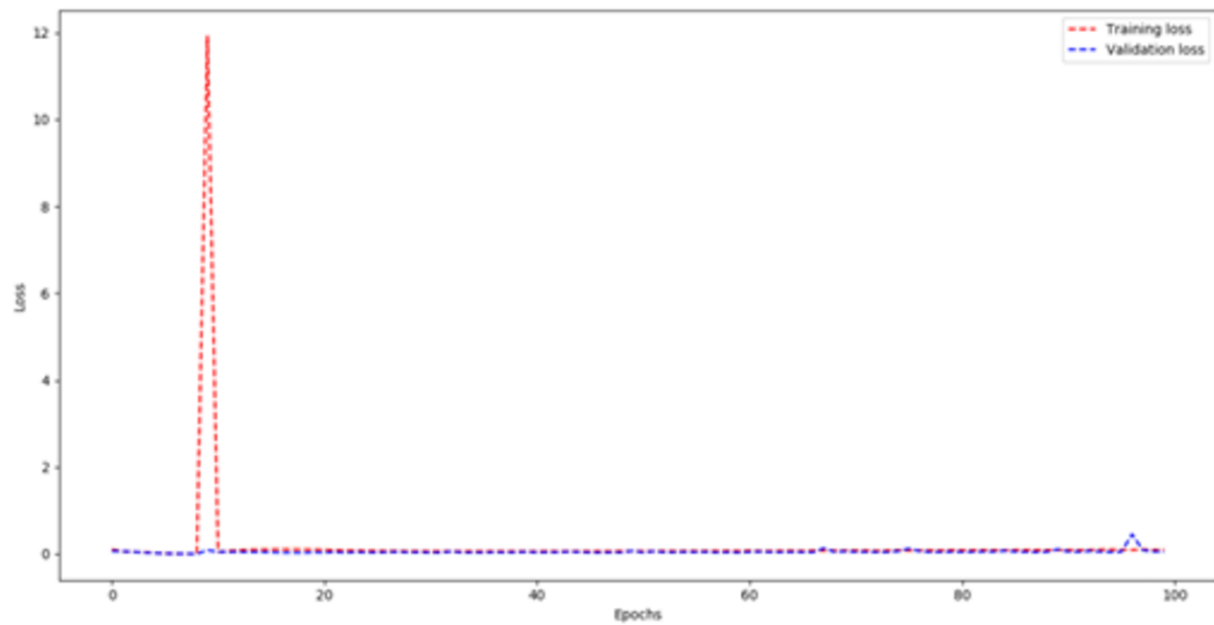


Fig.2. Bebop drone model training loss and validation loss curve

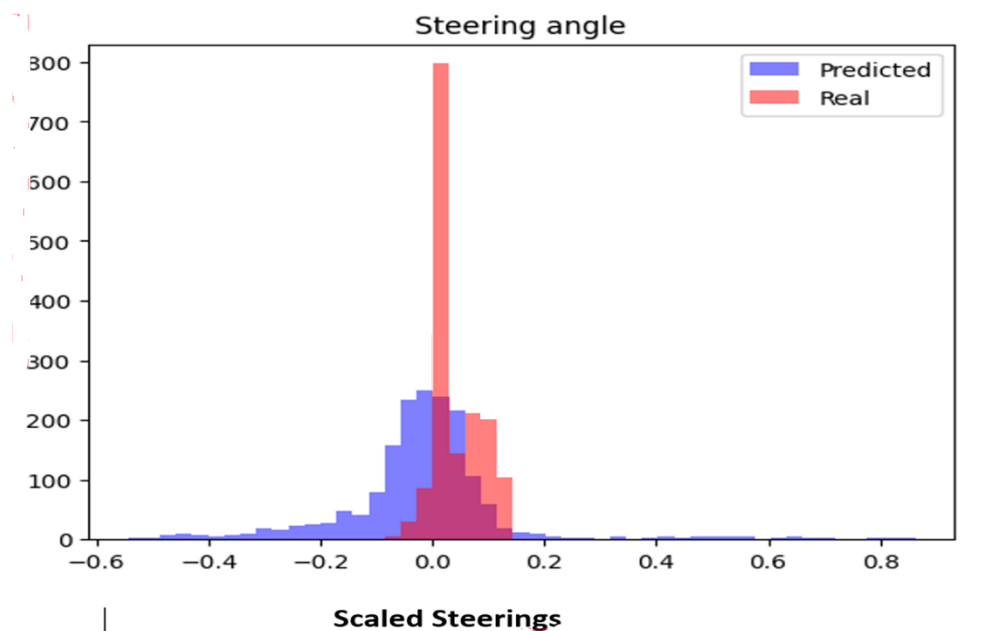


Fig.3. Steering angle probability density function

The points on the x-axis are the observations, and the y-axis is the likelihood of each observation. Observations around 0 are the most common, and the ones around -0.4 and 0.4 are rare. Negative values are used to indicate a left turn, while a positive value indicates a right turn.

The UAV model's probability density function's steering angles will control the UAV the same way the Gaussian Distribution steering angles do.

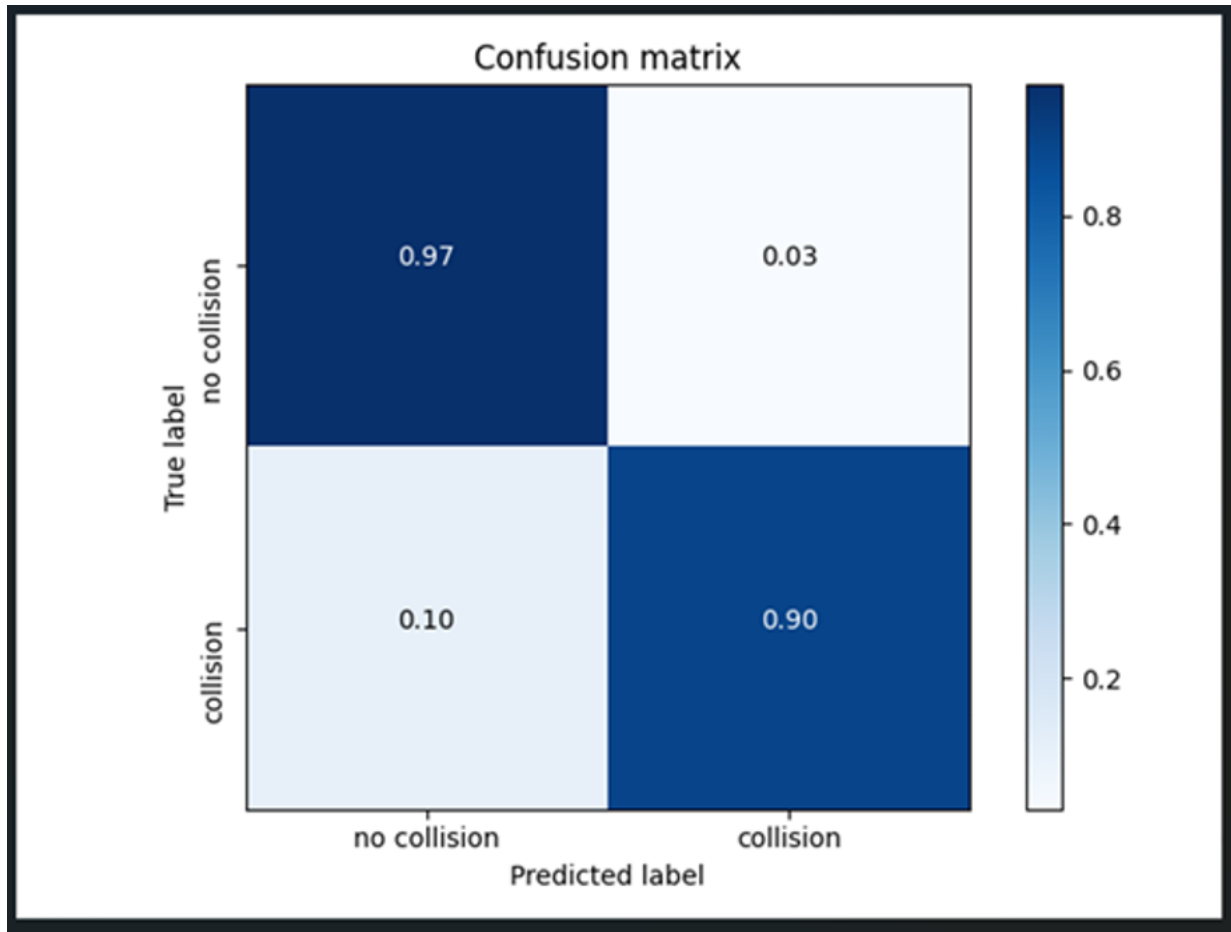


Fig.4. Bebop drone Model confusion matrix

Model	EVA	RMSE	Avg. accuracy	F-1 score	Num. Layers
Drone Model	0.865	0.181	95.3%	0.895	8

Fig.5. Drone model evaluation table

Implementation of Kalman Filters

The implementation of Kalman filters in this project played a crucial role in enhancing the accuracy and reliability of drone navigation and target tracking. One of the key functions of the Kalman Filter is in target pursuit, where it predicts and smooths the position of the target, such as in face detection. By considering the last known position and velocity of the target, it continuously updates its estimates with new sensor data, ensuring precise tracking and improving pursuit accuracy. This capability allows drones to stay on course when tracking a moving object, even when faced with noisy measurements.

Kalman filters also significantly mitigate the impact of low-light conditions and noisy sensor data. In the context of steering, the filter predicts the angle and refines it based on incoming measurements, reducing inconsistencies in navigation. When applied to face recognition, it helps compensate for challenges such as low resolution or lighting variations, ensuring that the system maintains a high level of accuracy even in adverse conditions. By filtering out errors and spikes caused by noisy data, the Kalman filter ensures that the drone can operate smoothly in unpredictable environments.

Additionally, Kalman filters contribute to improved navigation accuracy by maintaining stable steering angles and target positions. Their predictive adjustments allow the drone to make real-time corrections, promoting steady flight and reducing the risk of collisions. This functionality is particularly valuable for drones operating in complex environments, where precision and adaptability are essential for successful missions. Through these enhancements, the integration of Kalman filters strengthens the overall capabilities of the drone system, making it more efficient and reliable in dynamic operational scenarios.

Controlling the Drone

- The outputs of the model are used to command the UAV to move on a plane with forward velocity v_k and steering angle θ_k .

$$v_k = (1 - \alpha)v_{k-1} + \alpha(1 - p_t)V_{max}$$

- When probability of collision $p_t = 0$, forward velocity = V_{max}
- When probability of collision $p_t = 1$, forward velocity = 0
- Steering angle is calculated by mapping the predicted scaled steering s_k into a rotation around the z-axis.

$$\theta_k = (1 - \beta)\theta_{k-1} + \beta \frac{\pi}{2} s_k$$

- α and β perform low-pass filtering on the velocity and steering angle to provide smooth, continuous inputs.
- The paper uses $\alpha = 0.7$ and $\beta = 0.5$.

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Fig.6. Drone model implementation Results

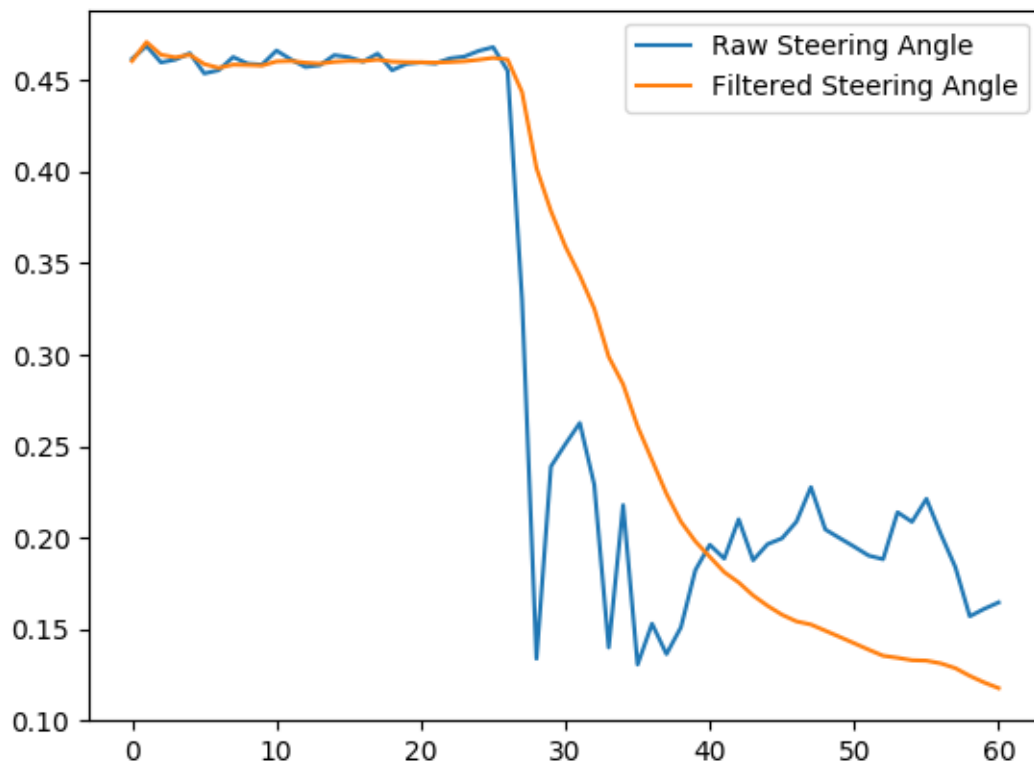


Fig. 7. Raw steering-angle measurements (blue) contrasted with the filtered steering-angle signal (orange), illustrating the reduction of high-frequency noise after signal smoothing.

Simulating tracking of a person using Latitude and Longitude

Simulating Tracking Process without using Kalman filters

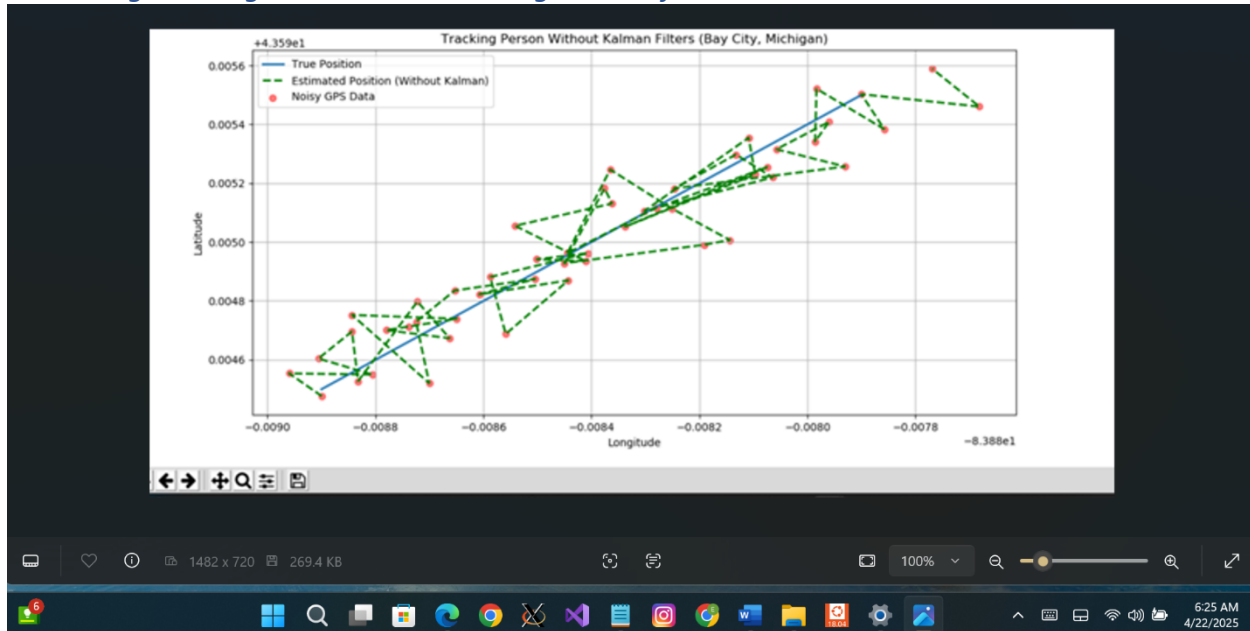


Fig.8. Simulating Tracking Process without using Kalman filters in Bay City, Michigan

True Positions: Simulates the actual movement of the person.

Noisy Positions: Adds noise to simulate real-world GPS inaccuracies.

Direct Tracking: Uses the noisy GPS data directly as the estimated positions without applying any Kalman filter.

Simulating Tracking Process with Kalman filters

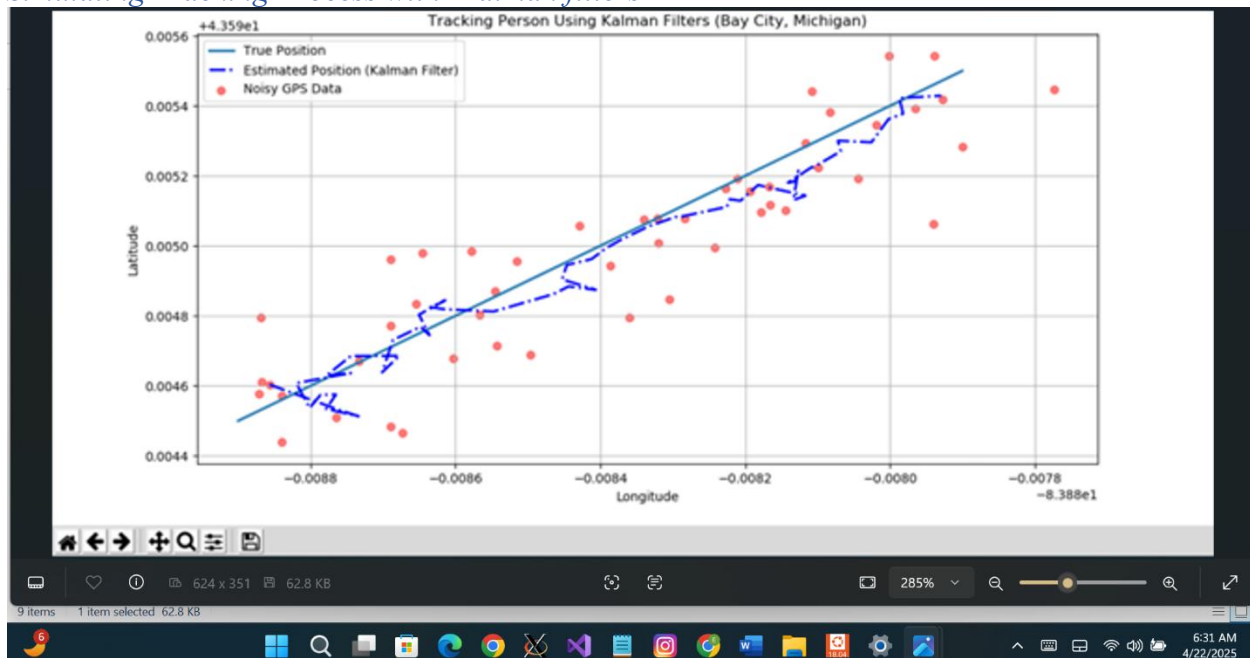


Fig.9. Tracking a person using Kalman filters in Bay City, Michigan

True Position Line: Represents the actual path taken by the person (green line).

Noisy GPS Data Points: Red dots scattered around the true path, representing the inaccuracies in the GPS data.

Estimated Position Line (Kalman Filter): Blue dashed line, representing the path estimated by the Kalman filter, which should be smoother and closer to the true path.

By comparing the two graphs, one without Kalman filter and one with Kalman filters it is evident that direct tracking without Kalman filters is more erratic and less accurate due to the noisy and that the Kalman filter reduces noise and provides a more accurate estimate of the true path

Metric	Without Kalman Filters	With Kalman Filters
MSE	1.08×10^{-8}	3.93×10^{-9}
RMSE	1.04×10^{-4}	6.27×10^{-5}
Mean Average Error (MAE)	9.00×10^{-5}	8.00×10^{-5}

Fig.10. evaluation results

The lower MAE value when using Kalman filters indicates that the Kalman filter significantly reduces the average error, providing more accurate tracking compared to using raw noisy data

Implementation Results: Integrating Dlib and GANs for Advanced Face Recognition

The integration of Dlib and Generative Adversarial Networks (GANs) has propelled advancements in face recognition systems, especially under challenging conditions like low-light or thermal environments. This comprehensive system enhances the drone's ability to perform accurate and reliable face recognition under challenging conditions.

Test Results of Implementing Dlib

Result of Face Recognition in the night time when the room light is turned on

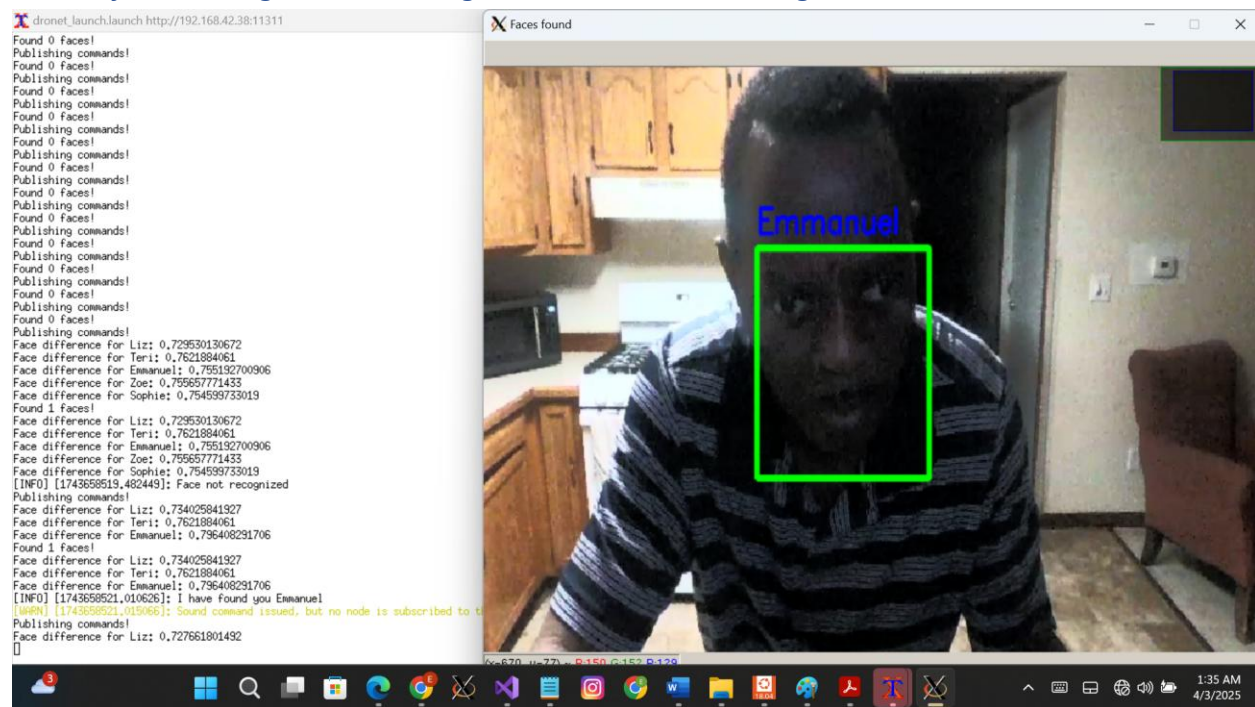


Fig. 11. Night-time scenario with the room light turned on.

Training Thermal-to-Visible GAN model

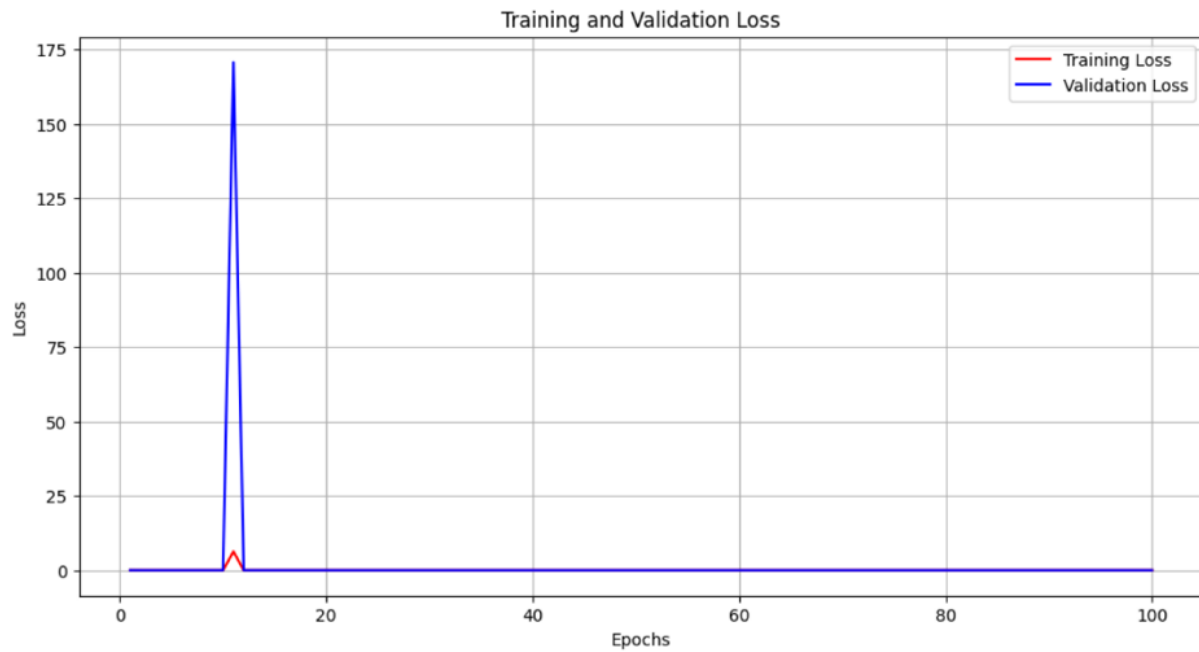
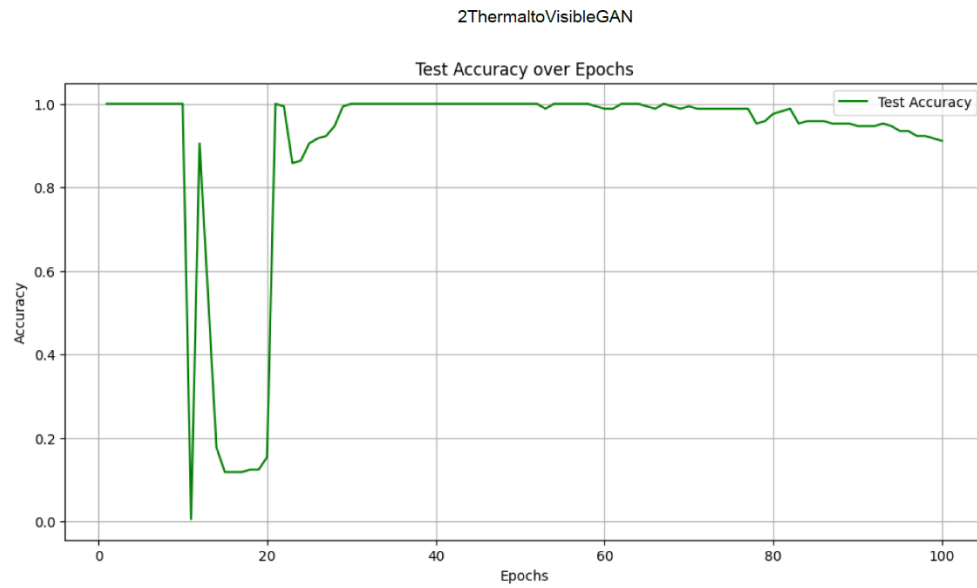


Fig.12. Training and validation loss over 100 epochs.

Discriminator Accuracy is: 0.9112



Training Completed!

Fig. 13. Test accuracy of the Thermal-to-Visible GAN across 100 epochs.

Test Results of Implementing Dlib and Thermal-to-Visible GAN Model

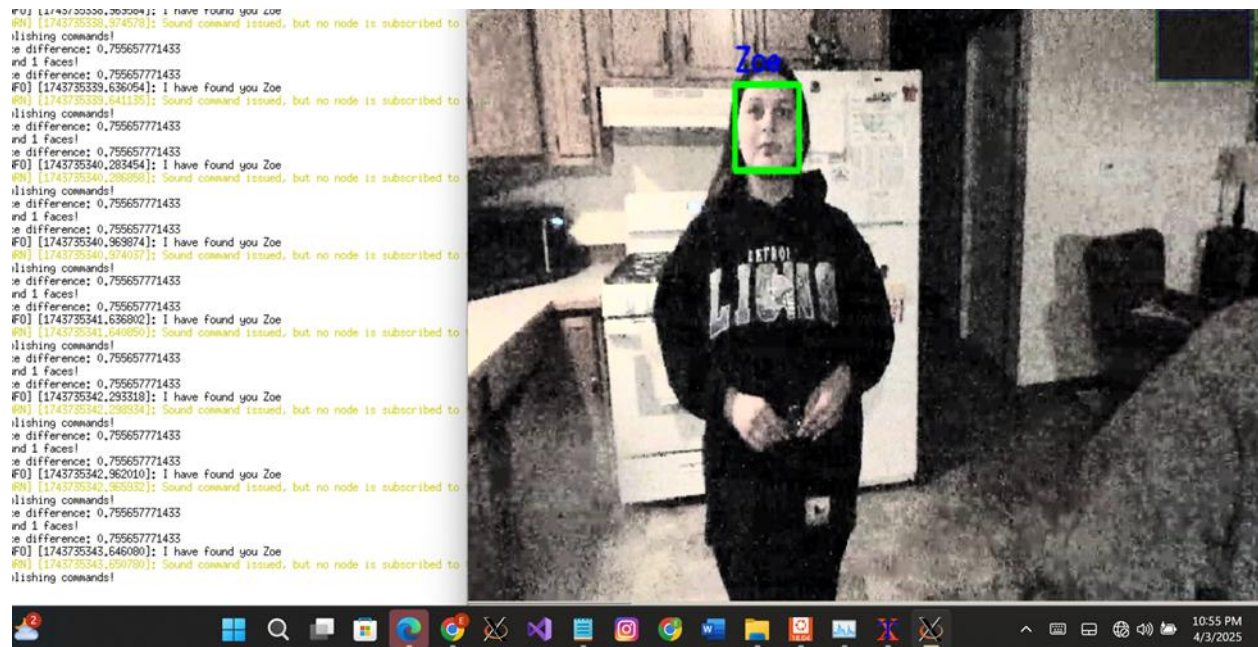


Fig.14. The Bebot drone's advanced face-recognition system demonstrated reliable performance across diverse subjects; the model successfully recognized Zoe.

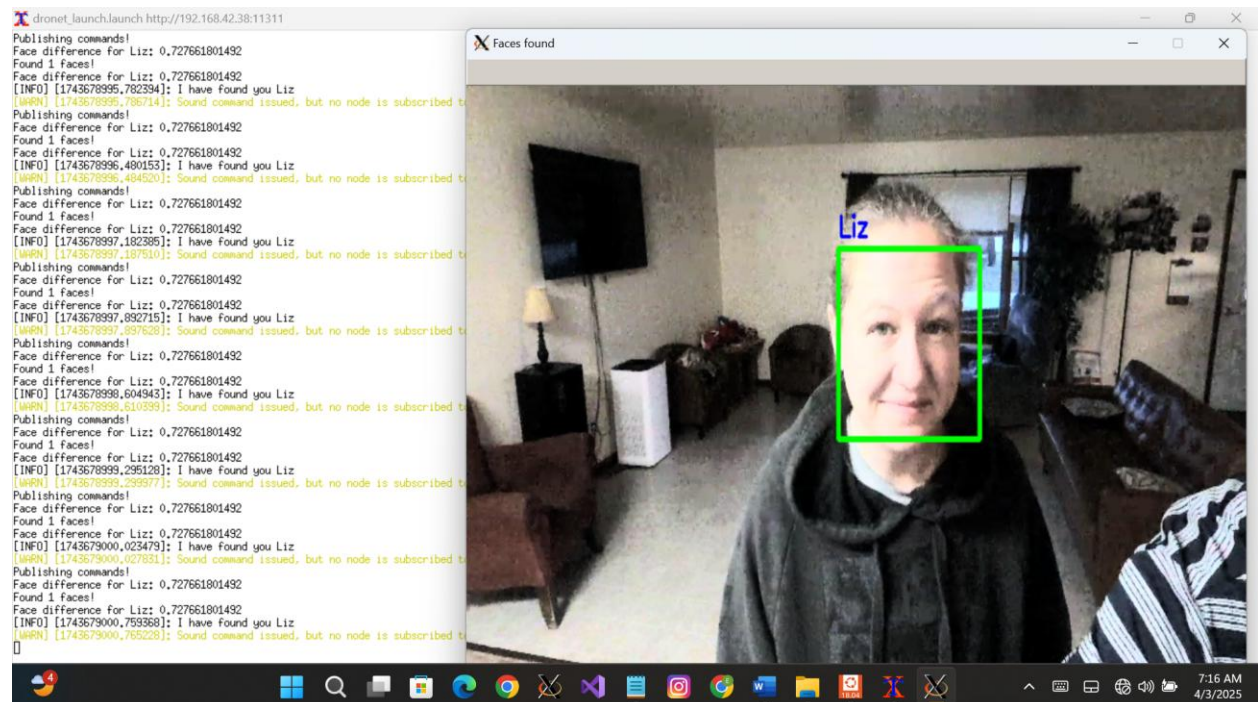


Fig.15. The Bebot drone's face-recognition system also correctly identified Liz, further confirming the robustness of the Thermal-to-Visible GAN pipeline.

The implementation of Thermal GANs enabled the conversion of thermal images into visible-light representations, significantly improving the accuracy of face recognition systems. This enhancement allows drones to perform reliable detection even in nighttime conditions.

Face Recognition in the Dark: Results of Thermal GAN Implementation

Night Time: The light in the room is turned off and the room is dark

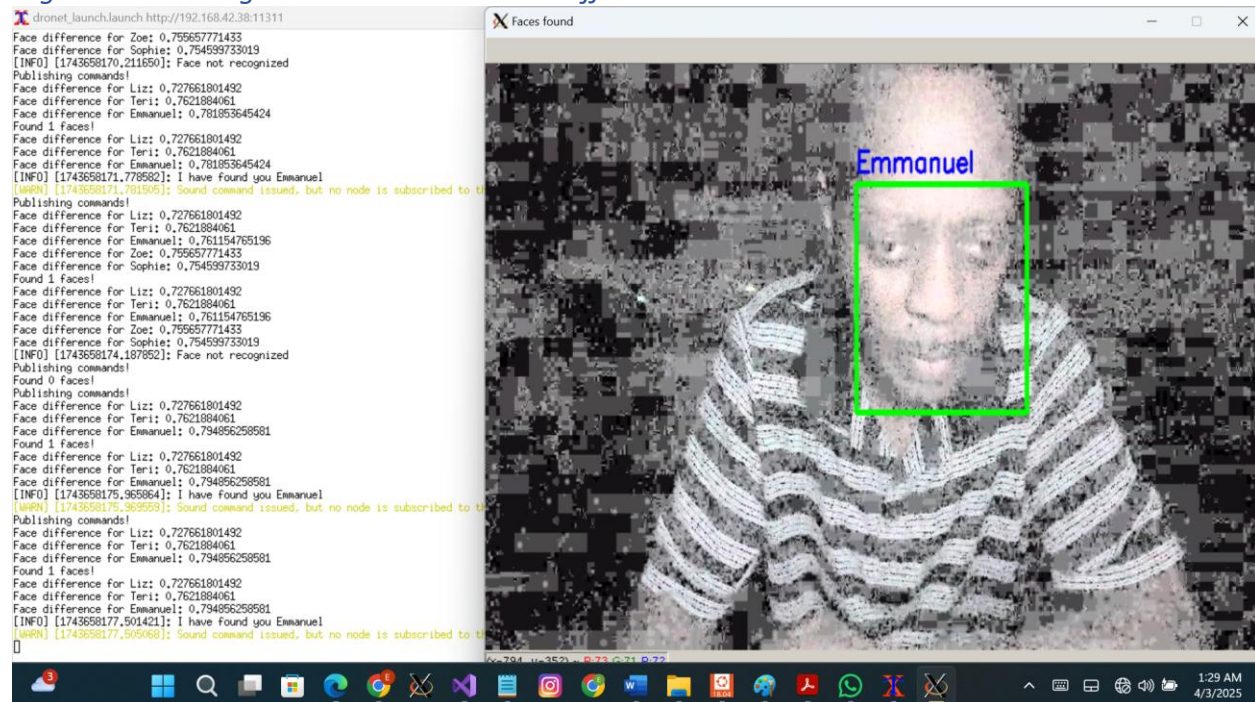


Fig. 16. Night-time scenario with the room light turned off, resulting in a fully dark environment. The system successfully performed face recognition under these low-illumination conditions.

Siamese Network GAN model training

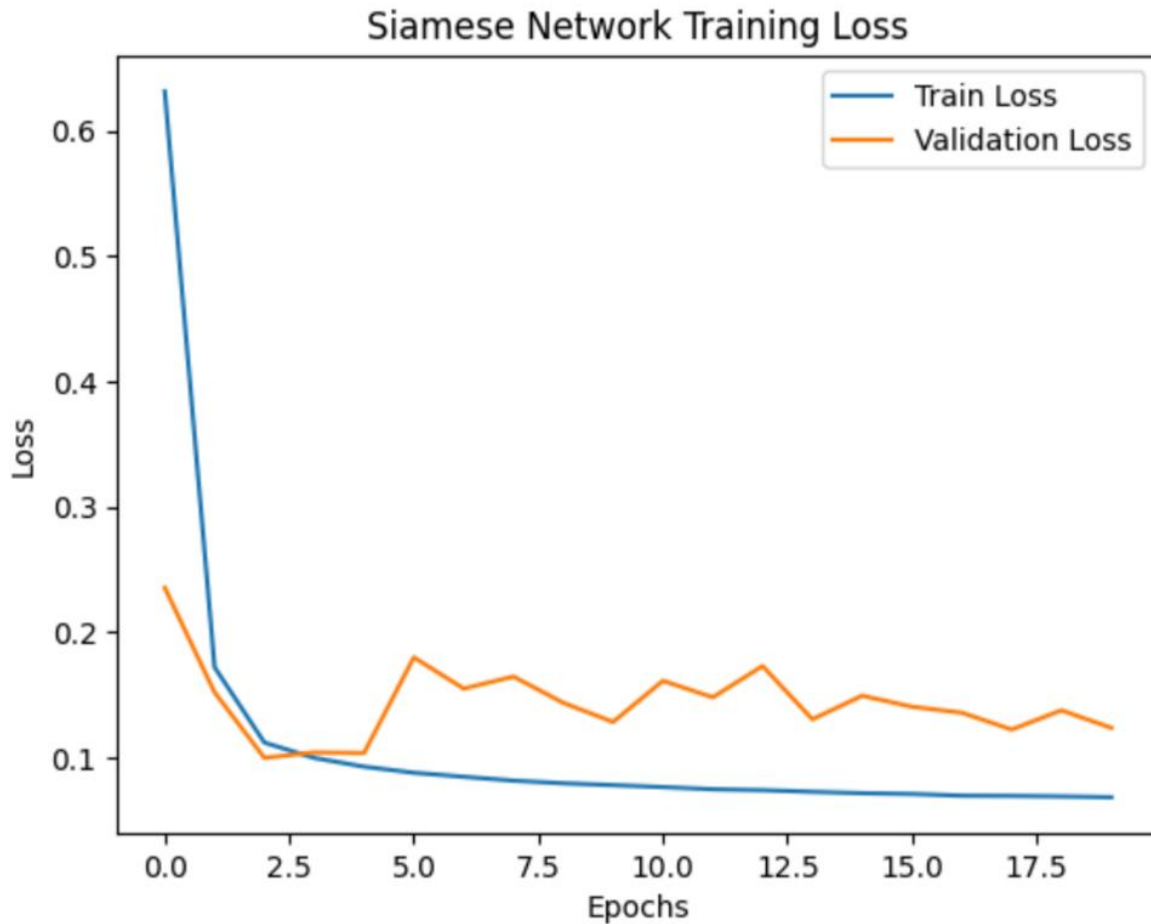


Fig.17. The Siamese Network achieved a validation accuracy of 84.65%, demonstrating strong verification performance across paired facial images. The model reliably distinguished between similar and dissimilar pairs using a learned embedding space and a 0.5 similarity threshold. This level of accuracy indicates that the network successfully captured identity-preserving features despite variations in lighting, pose, and modality. These results confirm the suitability of the Siamese architecture for real-time identity verification within the Bebop drone's face-recognition pipeline.

Siamese Network GAN Implementation Results

The Siamese network GAN (SiGAN) architecture improved face verification tasks under challenging conditions like low-light or thermal imaging scenarios. This system demonstrated superior accuracy in classifying image pairs and preserving identity integrity in reconstructed faces.



Fig.18. A dissimilar-pair classification was produced when real-time Image A of Emmanuel was compared with Image B of Teri.

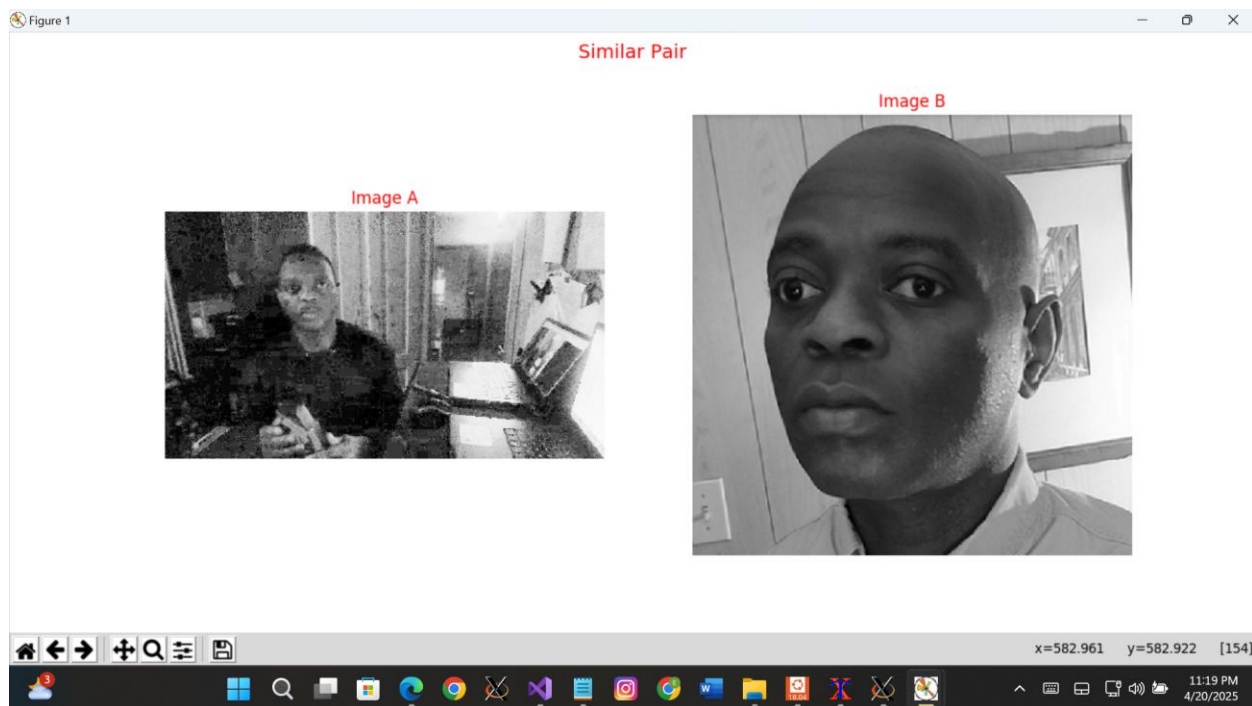


Fig.19. A similar-pair classification was produced when real-time Image A of Emmanuel was compared with his archived reference Image B.

Conclusion

This project represents a significant advancement in drone navigation and target tracking by integrating Kalman filters for precise state estimation and addressing challenges such as low-light conditions and noisy sensor data. Additionally, the incorporation of advanced face recognition techniques, including Thermal-to-Visible Generative Adversarial Networks (GANs) and Siamese Network GANs (SiGAN), has proven pivotal in improving face detection and recognition accuracy under adverse conditions. These technologies enable drones to accurately pursue and identify targets even in challenging environments.

Future Works

Building on the accomplishments of this project, future research could focus on further leveraging advanced GAN-based face recognition techniques. Exploring more sophisticated architectures for GANs and Siamese Networks may enhance facial recognition accuracy. Additionally, integrating multi-sensor data fusion, including LiDAR and infrared imaging, could provide a more holistic understanding of the environment. Another promising direction is the development of collaborative multi-drone systems utilizing GAN-enhanced communication for coordinated operations.

References

- [1] K. J. P. Fernando, “Enhancing Perception for Autonomous Driving Systems with Dual-Vision Input CycleGAN for Night-to-Day Image Translation,” 2018.
- [2] Z. Wang, Z. Chen, and F. Wu, “Thermal-to-Visible Facial Image Translation Using GANs,” 2022.
- [3] A. Loquercio, A. I. Maqueda, C. R. del-Blanco, and D. Scaramuzza, “DroNet: Learning to Fly by Driving,” IEEE Robotics and Automation Letters, 2018.
- [4] H. Manjunatha, A. Pak, D. Filev, and P. Tsiotras, “KARNet: Kalman Filter Augmented Recurrent Neural Network for Learning World Models,” 2020.
- [5] S. Kim, I. Petrunin, and H.-S. Shin, “A Review of Kalman Filter with Artificial Intelligence Techniques,” 2021.
- [6] D. Poster et al., “A Large-Scale, Time-Synchronized Visible and Thermal Face Dataset,” 2021.
- [7] Anonymous, “An Adaptive Unscented Kalman Filter for Vehicle Dynamics,” 2019.

- [8] M. A. Tahir, I. Mir, and T. U. Islam, "Control Algorithms, Kalman Estimation, and Near-Actual Simulation for UAVs: State-of-Art Perspective," 2020.
- [9] H. W. Hu, L. Kang, and Z. Yu, "A Possibilistic Risk Assessment Framework for Unmanned Electric Vehicles," *Frontiers in Energy Research*, 2022.
- [10] G. DeRose et al., "Autonomously Steering Vehicles Along Unmarked Roads Using Low-Cost Sensing," *Vehicles*, 2023.
- [11] J. Hai et al., "A Visual Navigation System for UAV under Diverse Illumination Conditions," *Applied Artificial Intelligence*, 2021.